

Yield Estimation Module

This component serves as the core of the proposed system, designed to automate the process of fruit detection, counting, and yield estimation using computer vision techniques. The module operates in three distinct phases: object detection, double-counting mitigation, and mathematical yield calculation.

1) Data Acquisition and Preprocessing:

To ensure the robustness of the model, a custom dataset consisting of 2,500 images of Coconuts was compiled. The images were captured under varying environmental conditions, including different lighting intensities and angles, to simulate real-world agricultural scenarios. Data annotation was performed using Roboflow, bounding each visible fruit. To prevent overfitting and improve the model's generalization capabilities, data augmentation techniques such as random rotation, horizontal flipping, and brightness adjustments were applied during the training phase.

2) Object Detection Architecture:

For the task of accurate fruit detection, the YOLOv8 (You Only Look Once) deep learning architecture was employed. YOLO was selected due to its superior trade-off between inference speed and detection accuracy, which is critical for processing high-resolution images on mobile devices. The model was fine-tuned using the prepared dataset, optimizing the loss functions for bounding box regression and classification to accurately identify fruits even when partially occluded by leaves or branches.

3) Double-Counting Mitigation via Acquisition Protocol:

Since the system operates on static images rather than continuous video feeds, a strict image acquisition protocol was implemented to minimize the risk of double-counting fruits.

- **Dual-View Strategy:** To cover the maximum yield without overlapping fields of view, the user is restricted to capturing images from two diametrically opposite sides of the tree (180-degree separation).
- **User Guidance System:** The mobile application includes an interactive warning and guidance mechanism. Before image capture, a prompt explicitly instructs the user to select two distinct, non-overlapping angles. This procedural constraint ensures that the fruits captured in the first image are distinct from those in the second, effectively reducing duplicate counts manually.

Revised Yield Formula:

Consequently, the total count (N) is derived by summing the detections from these discrete views:

$$N = N\{\text{side_A}\} + N\{\text{side_B}\}$$

4) Yield Estimation Formulation:

The final estimated yield is derived by mapping the total count of detected fruits to a mass value. The system utilizes a calibrated mathematical formula that incorporates the average weight of a specific fruit variety and a biological correction factor to account for fruits that are completely hidden from the camera's view.

The yield (Y_{est}) is calculated as follows:

$$Y_{\text{est}} = N * W_{\text{avg}} * C_{\text{f}}$$

Where:

- N represents the total unique count of fruits obtained from the dual-view counting process.
- W_{avg} denotes the average weight of a single fruit (derived from historical agricultural data).
- C_{f} is the correction factor (e.g., 1.1) applied to compensate for occlusion levels inherent in the specific tree canopy structure.